

A disproof of Conjecture 00000001354

Square-root-phase prime sums admit no uniform bound $x^{1/4+\varepsilon}$

1 The conjecture

Conjecture 00000001354 reads (English version):

Definition: The square-phase prime sum is $\sum_{p \leq x} e(\sqrt{p})$. *Conjecture:* $\sum_{p \leq x} e(\sqrt{p}) \ll x^{1/4+\varepsilon}$; and the same upper bound holds uniformly after replacing $\sqrt{p}$ by $\alpha\sqrt{p}$ for any fixed $0 < \alpha \leq 1$ – square-root cancellation for prime sums with square-root phases, uniformly.

The Chinese version (in `conjectures/00000001354.md`) says, in transliteration, “ba $\sqrt{p}$ huan wei $\alpha\sqrt{p}$ (renyi guding $0 < \alpha \leq 1$) hou tong yi shangjie *yizhi chengli*, ji pingfanggen xiangwei de sushu he you *junyun* pingfanggen dixiao”: “after replacing $\sqrt{p}$ by $\alpha\sqrt{p}$ (any fixed $0 < \alpha \leq 1$) the same upper bound *holds uniformly*, i.e. prime sums with square-root phases have *uniform* square-root cancellation”.

Write $e(t) = \exp(2\pi it)$ and $S_\alpha(x) = \sum_{p \leq x} e(\alpha\sqrt{p})$, the sum over primes $p \leq x$. The conjecture is the conjunction of

- (1) for every $\varepsilon > 0$: $S_1(x) \ll_\varepsilon x^{1/4+\varepsilon}$;
- (2) for every $\varepsilon > 0$: $S_\alpha(x) \ll_\varepsilon x^{1/4+\varepsilon}$ uniformly in $0 < \alpha \leq 1$.

Result. Clause (2) is false for every $0 < \varepsilon < 3/4$, so the conjecture is false. Taking $\alpha = 1/(8\sqrt{x})$ makes every phase lie in $[0, \pi/4]$, so $|S_\alpha(x)| \geq \cos(\pi/4) \pi(x)$, and Chebyshev’s lower bound for $\pi(x)$ beats $Cx^{1/4+\varepsilon}$ for every C . Everything is formalized in Lean 4 with Mathlib (Section 5).

2 Reading

- (a) **Uniformity.** A bound $f(x, \alpha) \ll g(x)$ holds *uniformly* in $\alpha \in A$ if one implied constant (and one range of x) serves every $\alpha \in A$; when the constant may depend on a parameter, this is indicated by a subscript (Section 3). We formalize clause (2) as: for every $\varepsilon > 0$ there are C and X , independent of α , with $|S_\alpha(x)| \leq Cx^{1/4+\varepsilon}$ for all $\alpha \in (0, 1]$ and all $x \geq X$. Allowing a threshold X only weakens the claim compared with the usual convention ($x \geq 2$), so the refutation covers both.
- (b) **Why this is the literal reading.** The Chinese text says the bound *yizhi chengli* (holds uniformly), and then restates the claim as *junyun* (uniform) square-root cancellation. *Yizhi* is the standard Chinese term for “uniform(ly)” in analysis (*yizhi shoulian*, uniform convergence; *yizhi lianxu*, uniform continuity). The English says “uniformly” twice. If the constant could depend on α , both words would carry no content, because “the same upper bound” already fixes the exponent $1/4 + \varepsilon$. The phrase “for any fixed $0 < \alpha \leq 1$ ” names the parameter range over which uniformity is asserted, as in the standard phrasing “for fixed α ..., uniformly in α ”.

- (c) **What is not claimed.** We do not refute clause (1) on its own, nor the non-uniform reading in which C may depend on α . Both are open problems about primes in short intervals, and nothing below depends on them. The conjecture, read with its uniform clause, is false because that clause is false.

3 Definitions and sources

Definition 1. $e(t) = \exp(2\pi it)$ for real t ; $\pi(x)$ is the number of primes $p \leq x$; for real α, x , $S_\alpha(x) = \sum_{p \leq x, p \text{ prime}} e(\alpha\sqrt{p})$. The conjecture's square-phase prime sum is $S_1(x)$.

Sources (quotations verbatim):

- Wikipedia, *Big O notation*, https://en.wikipedia.org/wiki/Big_O_notation, on bounds in several variables: “In all of these examples, the bound is uniform in both variables.” On a parameter-dependent constant, marked by a subscript: “Here the implied constant depends on the size of the domain.”
- Chebyshev's lower bound, as formalized in Mathlib (file `Mathlib/NumberTheory/Chebyshev.lean`, theorem `Chebyshev.pi_ge'`): for real $x > 1$,

$$\pi(x) \geq \frac{(x-1)\log 2 - \log(x+2)}{\log x}.$$

4 The proof

Lemma 2. For $x > 0$ and $\alpha = 1/(8\sqrt{x})$: $\operatorname{Re} S_\alpha(x) \geq \frac{\sqrt{2}}{2} \pi(x)$.

Proof. $\operatorname{Re} e(t) = \cos(2\pi t)$. For a prime $p \leq x$, the phase is $2\pi\alpha\sqrt{p} = \frac{\pi}{4} \cdot \frac{\sqrt{p}}{\sqrt{x}} \in [0, \pi/4]$. Since $\cos$ decreases on $[0, \pi]$, $\cos(2\pi\alpha\sqrt{p}) \geq \cos(\pi/4) = \sqrt{2}/2$. Summing over the $\pi(x)$ primes $p \leq x$ gives the claim. $\square$

Lemma 3. Let $0 < \theta < 1$. For all real K and X there is $x \geq \max(X, 2)$ with $Kx^\theta < \pi(x)$.

Proof. Let $\eta = (1-\theta)/2 \in (0, 1/2)$ and choose $x \geq \max(X, 2)$ with $y := x^\eta > 4(|K|+2)/\eta$ (possible since $x^\eta \rightarrow \infty$). Put $z = x^{1-\eta}$, so $x = zy$, $x^\theta y = z$, and $y \leq z$ (as $\eta \leq 1-\eta$ and $x \geq 1$). We use $\log x \leq x^\eta/\eta = y/\eta$, $\log(x+2) \leq 2\log x$ (since $x+2 \leq x^2$), $\log 2 > 1/2$ and hence $(x-1)\log 2 \geq x/4$. Chebyshev's bound gives $(x-1)\log 2 - \log(x+2) \leq \pi(x)\log x$. Multiplying by η and using the inequalities above,

$$\eta x/4 - 2y \leq \eta((x-1)\log 2 - \log(x+2)) \leq \eta\pi(x)\log x \leq \pi(x)y.$$

Since $\eta y/4 > |K| + 2$, we get $\eta x/4 = (\eta y/4)z > (|K|+2)z \geq Kz + 2y$, and so $Kx^\theta y = Kz < \eta x/4 - 2y \leq \pi(x)y$. Dividing by $y > 0$ gives $Kx^\theta < \pi(x)$. $\square$

Theorem 4. For every $0 < \varepsilon < 3/4$ and all C, X there are $\alpha \in (0, 1]$ and $x \geq X$ with $|S_\alpha(x)| > Cx^{1/4+\varepsilon}$. Hence clause (2) is false, and so is the conjecture.

Proof. Apply Lemma 3 with $\theta = 1/4 + \varepsilon$ and $K = 2C$ to get $x \geq \max(X, 2)$ with $2Cx^\theta < \pi(x)$. Put $\alpha = 1/(8\sqrt{x})$. Then $0 < \alpha \leq 1$ because $\sqrt{x} \geq 1$, and by Lemma 2

$$|S_\alpha(x)| \geq \operatorname{Re} S_\alpha(x) \geq \frac{\sqrt{2}}{2} \pi(x) \geq \frac{1}{2} \pi(x) > Cx^{1/4+\varepsilon}.$$

(For $\varepsilon \geq 3/4$ the bound is trivial, since $|S_\alpha(x)| \leq \pi(x) \leq x$.) $\square$

5 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). `Conjecture1354.lean` imports the single source file `Conjecture1354/Basic.lean` (179 lines, namespace `Conjecture1354`, only import `Mathlib`).

Definitions.

- `e t := Complex.exp (2 * π * Complex.I * t).`
- `S α x := ∑ p ∈ Nat.primesLE [x]+, e (α * √(p : ℝ));` Mathlib's `Nat.primesLE n` is the finset of primes $\leq n$, so this is $S_\alpha(x)$.
- `AlphaOneClause: ∀ ε > 0, ∃ C X, ∀ x ≥ X, ||S 1 x|| ≤ C * x ^ (1/4 + ε)` (clause (1)).
- `UniformBound ε: ∃ C X, ∀ α ∈ Set.Ioc 0 1, ∀ x ≥ X, ||S α x|| ≤ C * x ^ (1/4 + ε).`
- `UniformClause := ∀ ε > 0, UniformBound ε` (clause (2)).
- `Conjecture := AlphaOneClause ∧ UniformClause.`

Theorems.

- `e_re : (e t).re = Real.cos (2 * π * t).`
- `re_S_ge (hx : 0 < x) : √2 / 2 * π([x]+) ≤ (S (1 / (8 * √x)) x).re` (Lemma 2; π is `Nat.primeCounting`).
- `exists_large (hθ0 : 0 < θ) (hθ1 : θ < 1) (K X : ℝ) : ∃ x, X ≤ x ∧ 2 ≤ x ∧ K * x ^ θ < π([x]+)` (Lemma 3, from `Chebyshev.pi_ge'` and `Real.log_le_rpow_div`).
- `not_uniformBound (hε0 : 0 < ε) (hε : ε < 3 / 4) : ¬ UniformBound ε` (Theorem 4).
- `not_uniformClause : ¬ UniformClause.`
- The main theorem

`conjecture1354_false : ¬ Conjecture.`

Axiom audit. There is no sorry, no `native_decide` and no added axiom. `#print axioms reports`

`'Conjecture1354.conjecture1354_false'` depends on axioms:
`[propext, Classical.choice, Quot.sound]`

and the same for `not_uniformBound`. The file also elaborates with `-DwarningAsError=true`. To check:

```
cd lean
lake exe cache get
lake build
```